EE 115 Homework 4

1) To generate a conventional AM signal, we can use a (memoryless) nonlinear device followed by a bandpass filter. Assume that the input-output relationship of the nonlinear device is $x_{out} = \alpha x_{in}^2 - \beta x_{in}$, the input to the nonlinear device is $x_{in}(t) = m(t) + \cos(2\pi f_c t)$ where $f_c = 4B$ and B is the bandwidth of $m(t)$, and $m(t) = 2\mathtt{sinc}(3t)$ where t is in millisecond.
   a) **(20 points)** Sketch the spectrum of the output $x_{out}(t)$ of the nonlinear device.
   b) **(20 points)** Determine the frequency response $H(f)$ of the bandpass filter so that its output is a conventional AM signal $u_{AM}(t)$.
   c) **(10 points)** Determine the sufficient and necessary constraint on the positive parameters α and β so that the negative $-m(t)$ of the message can be extracted out from $u_{AM}(t)$ using an envelope detector and a DC blocker.
2) Consider a DSB-SC signal $u_1(t) = m(t)\cos(200\pi t)$ followed by an ideal bandpass filter with the frequency response $H(f)$. Assume that $m(t)$ has the spectrum

$$M(f) = \begin{cases} 10 + f, & -10 \le f \le 0 \\ 10 - f, & 0 \le f \le 10 \\ 0, & \mathtt{otherwise} \end{cases} \tag{1}$$

   and $H(f) = 1$ if $|f - 105| \le 5$ or $|f + 105| \le 5$, and $H(f) = 0$ otherwise. In this problem, t is in second and f is in Hertz.
   a) **(20 points)** Sketch the spectrum of the output $u_2(t)$ of the bandpass filter.
   b) **(20 points)** Determine $u_2(t)$ in the following form:

$$u_2(t) = u_c(t)\cos(2\pi f_c t) - u_s(t)\sin(2\pi f_c t) \tag{2}$$

   where $f_c = 100$. Specify $u_c(t)$ and $u_s(t)$.
   c) **(10 points)** Should $u_s(t)$ be the Hilbert transform of $u_c(t)$? Explain.